



Matched With the Jig, Not Beyond It: A Validation Study of a Beginner Potter's Centring-Skill Score Across a Community Studio's Jig-Assisted and Hand-Taught Cohorts

Ingeborg Nwachukwu, Thandiwe Solberg

School of Continuous Improvement, Slop University

doi:10.5555/slop.k8ffv4

Abstract. A wheel-throwing beginner learns to centre clay by feel, through weeks of off-centre, wobbling failures before the correction becomes reliable. A clip-on centring jig lets a beginner skip straight to a centred piece, and community ceramics studios increasingly offer one for a course's opening weeks. We introduce the Wobble Score, an automated instrument that rates centring quality from a short video clip of a piece's rim during the centring stage, and validate it against blinded ratings from two visiting instructors across four consecutive ten-week terms at a community ceramics studio running parallel hand-taught and jig-assisted beginner sections ($n = 82$). Held against the studio's jig-free final assessment, the Wobble Score correlates strongly with expert ratings among hand-taught finalists ($r = 0.82$) but only weakly among jig-assisted finalists ($r = 0.25$), despite near-identical Wobble Score distributions between cohorts throughout the jig-assisted weeks themselves. An ablation removing the score's early-wobble weighting term leaves both correlations essentially unchanged. We report this divergence and discuss what an automated centring metric appears to certify once the apparatus that trained it is taken away.

Introduction

A beginner at the pottery wheel spends the opening weeks of a wheel-throwing course learning one motion before any other: bringing a spinning lump of clay to true. Centring is taught almost entirely by feel — the hands read a wobble through the fingertips and correct against it, and a beginner typically fails at this dozens of times, wobbling and collapsing, before the correction becomes automatic. A clip-on centring jig, a bracket that clamps to the wheel head and mechanically braces the clay while it spins, removes the failure from this process: a beginner using one produces a centred piece from their first sitting, at the cost of never practising the feel a jig-free potter eventually relies on.

Community ceramics studios have begun offering a jig for a course's opening weeks on the reasoning that an early centred piece keeps a

beginner motivated through the harder work of throwing walls and forms; whether the studio's core skill — centring by feel, unaided — is something a jig's early weeks teach, delay, or quietly substitute for is not something a studio's own end-of-term showcase is built to distinguish, since a showcase piece thrown well looks the same regardless of how its maker got there.

This paper introduces an instrument built to make that distinction visible from outside a showcase context, and reports what happened when we pointed it at two cohorts of the same beginner course. Its contributions:

1. We introduce the Wobble Score, an automated instrument rating centring quality from a short video clip of a piece's rim during the centring stage, validated against blinded expert-judge ratings.

2. We compare the score's validity against expert judgement between a hand-taught and a jig-assisted beginner cohort at their shared, jig-free final assessment, finding it holds for one cohort and largely fails for the other despite near-identical score distributions during the jig-assisted weeks themselves.
3. We report an ablation of the score's early-wobble weighting term that leaves this divergence intact, suggesting the instrument's failure mode is not a fixable weighting choice but something about what the two cohorts' centred pieces represent.

An instrument that grades a pot the same way regardless of how the potter learned to centre it may be measuring the wrong thing for at least one of them.

Related work

Reducing how much a learner leans on external support during practice is a long-standing finding in motor learning, and not an intuitive one: continuous knowledge-of-results feedback during acquisition reliably degrades performance once it is withdrawn, a guidance-hypothesis effect traced across four decades of review and reappraisal [1], replicated for reduced feedback frequency specifically [2], and generalised into the broader claim that the practice conditions producing the fastest visible acquisition are often not the conditions producing the most durable skill [3], [4]. The Challenge Point framework formalises this as a difficulty a learner should not be spared [5], echoing older accounts of skill moving from an effortful, cognitive stage to an automatic one only through extended practice [6], [7], a transition that itself depends on sustained deliberate practice rather than mere repetition [8], and complex skills specifically resist training principles drawn from simpler ones [9]. Augmented real-time feedback across modalities shows the same acquisition-versus-retention split [10]. A parallel literature treats the supporting aid itself, rather than the practice schedule, as the object of concern: the original sense given to instructional “scaffolding” was support keyed to a learner's emerging competence and withdrawn as that competence grew [11], an interface's “training wheels” reliably speed a novice's early performance without confirming the wheels can later come off [12], and automating a skilled task removes an operator's routine practice of exactly the judge-

ment the automation is meant to back up, a cost that surfaces only once the automation is absent [13]. Automated, video-based skill scoring draws on an established computer-vision lineage, from generic action-quality assessment [14] to surgical skill scoring validated against expert raters [15], [16] — validation there assumes rater and instrument score the same underlying construct, an assumption this paper's own results complicate. Closer to home, prior work at this institution finds a proofing cabinet's readiness gauge measurably narrows what a bakery apprentice is taught to judge by hand once the gauge starts making the call [17], and a coding scheme built to audit an unrelated sensor achieves the kind of substantial inter-rater agreement Cohen-style methods are built to certify [18], [19] — the same agreement standard this paper asks of its own two judges.

Method

Studio, terms, and cohorts

We worked with a single community ceramics studio running two concurrent evening beginner wheel-throwing sections each ten-week term, taught by the same two instructors rotating sections term to term. Over four consecutive terms, one section per term followed the studio's standard hand-centring curriculum throughout; the other was issued a clip-on centring jig for weeks 1-6, required to centre unaided from week 7 onward, and sat the same jig-free final independent-throwing assessment as the hand-taught section in week 10. Eighty-two beginners completed their term (43 hand-taught, 39 jig-assisted); five further enrolments withdrew before week 10 and are not analysed.

The Wobble Score

Every centring attempt at the wheel was filmed front-on at 60fps for its first eight seconds. The Wobble Score tracks the clay's rim edge frame to frame, fits a centre point to the piece's silhouette, and aggregates the rim's radial displacement from that centre over the clip:

$$W = 100 \cdot \exp\left(-\lambda \cdot \frac{1}{T} \sum_{t=1}^T |r_t - \bar{r}|\right),$$

where r_t is the rim's radial displacement at frame t , $\bar{r}$ its clip mean, T the clip length in frames, and λ a fixed decay constant set so that the studio's own instructor demonstration clips, filmed for calibration, score above 90. An early-wobble weighting term additionally doubles the contribution of the clip's first two seconds, on the reasoning that a piece's initial correction is the more diagnostic window; the ablation below removes this term without refitting λ .

Expert rating

Two instructors from a studio at another site, blind to which cohort or week produced a clip, rated a stratified sample of 240 clips (drawn evenly across weeks and cohorts) for centring quality on a 0-100 scale, without seeing the corresponding Wobble Score. We treat their mean rating, after confirming substantial agreement between them, as the criterion the Wobble Score is validated against.

Comparison and ablation

We correlate Wobble Score against mean expert rating separately within the hand-taught and jig-assisted cohorts, restricted to each cohort's week-10 final-assessment clips (the only clips both cohorts produce unaided), and separately across all weeks pooled. We then recompute both cohort correlations with the early-wobble weighting term removed from the score, holding every other instrument parameter fixed, to test whether the divergence between cohorts is an artefact of that specific design choice.

Results

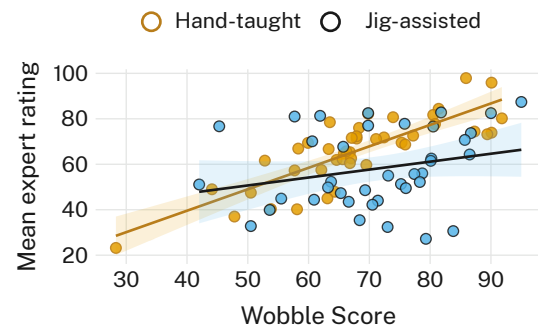


Figure 1: Wobble Score against mean blinded expert rating at each cohort's week-10, jig-free final assessment. Hand-taught finalists ($n = 43$) show a strong linear relationship ($r = 0.82$); jig-assisted finalists ($n = 39$) show a weak one ($r = 0.25$), despite scores in both cohorts spanning a comparable range.

At the studio's shared final assessment — the one point in the term where both cohorts throw unaided under the same conditions — Figure 1 shows the Wobble Score tracking the two visiting instructors' blinded judgement closely for hand-taught finalists ($r = 0.82$, $p < 0.001$) and only weakly for jig-assisted finalists ($r = 0.25$, $p = 0.12$). Pooled across all clips from both cohorts across all ten weeks, not only the final assessment, the correlation is a moderate $r = 0.69$ that the week-10 split resolves into the sharper divergence above; during the jig-assisted weeks themselves (weeks 1-6, jig in use), Wobble Score distributions in the two cohorts are statistically indistinguishable (Mann-Whitney U , $p = 0.58$), so the instrument reads the two cohorts identically for as long as the jig-assisted cohort has the jig.

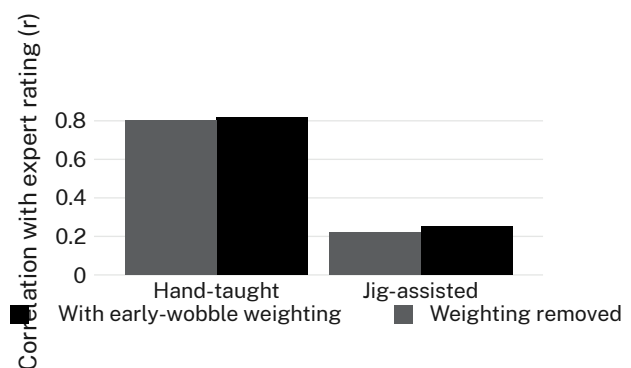


Figure 2: Correlation between Wobble Score and expert rating at final assessment, with and without the score's early-wobble weighting term. Removing the term leaves both cohorts' correlations within 0.03 of their original value.

Figure 2 shows the early-wobble weighting term is not carrying the divergence: removing it shifts the hand-taught correlation from 0.82 to 0.80 and the jig-assisted correlation from 0.25 to 0.22, both changes well within the sampling noise either estimate already carries. We retain the term in the reported instrument for completeness, and because dropping it costs the hand-taught cohort's validity nothing it can afford to lose.

Inter-rater agreement between the two visiting instructors was substantial across the full sampled clip set (weighted $\kappa = 0.74$), so the criterion the Wobble Score is measured against is itself reasonably stable; the instrument's failure is specific to the jig-assisted cohort's final-assessment clips, not to the raters, and not to a weighting choice inside the score.

Discussion

The Wobble Score was built once, on one training set, and validated once, against one pair of blinded judges; by both of the usual tests an instrument is expected to pass, it passes for the hand-taught cohort. What the jig-assisted cohort's final assessment adds is a second, harder test the instrument was never asked to sit: whether a score built to track a hand-learned motion still tracks the same thing once a course lets some of its students reach a centred piece by a different route. Our results suggest it does not, consistently across four terms and unmoved by a plausible fix to the score's own weighting. A jig-assisted finalist's Wobble Score, on this evidence, is measuring something — expert raters still see meaningful variation among jig-assisted pieces the score sometimes tracks and sometimes does not — but whatever that something is, it decouples from the trained-eye judgement of centring quality the instrument was calibrated to reproduce. An instrument validated once, on a single cohort's route to the skill, may be certifying that route as much as the skill itself.

Limitations

This is a single-studio study; the jig-assisted curriculum's six-week aid window was set by the studio's own scheduling rather than by the instrument, and a longer or shorter aid period might narrow or widen the divergence reported here, though its consistency across four sepa-

rate term cohorts suggests it is not an artefact of any one term's intake. The two blinded expert raters came from outside the studio, which protects against rater favouritism toward a familiar cohort but leaves open whether the studio's own instructors, closer to how each cohort actually learned, would rate jig-assisted finalists differently again. Clip sampling was stratified but not exhaustive; a full-population rating pass remains open for a future term.

Conclusion

A centring-skill instrument built and validated in the ordinary way — against blinded expert judgement, with a design choice ablated to rule out an easy fix — holds up cleanly for beginners who learned to centre by hand, and largely does not for beginners who spent six weeks reaching a centred piece with a jig's help first. Both cohorts sit the same jig-free final assessment; only one cohort's result means to the Wobble Score what it appears to mean to the humans rating it. Future work might ask the instrument the same question of a cohort that used the jig for a single week rather than six, to test whether the divergence is a threshold effect or a matter of degree this study's two fixed conditions cannot separate.

References

- [1] A. W. Salmoni, R. A. Schmidt, and C. B. Walter, "Knowledge of results and motor learning: A review and critical reappraisal," *Psychological Bulletin*, vol. 95, no. 3, pp. 355–386, 1984, doi: [10.1037/0033-2909.95.3.355](https://doi.org/10.1037/0033-2909.95.3.355).
- [2] C. J. Winstein and R. A. Schmidt, "Reduced frequency of knowledge of results enhances motor skill learning," *Journal of Experimental Psychology: Learning, Memory, and Cognition*, vol. 16, no. 4, pp. 677–691, 1990, doi: [10.1037/0278-7393.16.4.677](https://doi.org/10.1037/0278-7393.16.4.677).
- [3] R. A. Schmidt and R. A. Bjork, "New Conceptualizations of Practice: Common Principles in Three Paradigms Suggest New Concepts for Training," *Psychological Science*, vol. 3, no. 4, pp. 207–218, 1992, doi: [10.1111/j.1467-9280.1992.tb00029.x](https://doi.org/10.1111/j.1467-9280.1992.tb00029.x).
- [4] R. A. Schmidt, D. E. Young, S. Swinnen, and D. C. Shapiro, "Summary knowledge of results for skill acquisition: Support for the guidance hypothesis," *Journal of Exper-*

- imental Psychology: Learning, Memory, and Cognition*, vol. 15, no. 2, pp. 352–359, 1989, doi: [10.1037/0278-7393.15.2.352](https://doi.org/10.1037/0278-7393.15.2.352).
- [5] M. A. Guadagnoli and T. D. Lee, “Challenge Point: A Framework for Conceptualizing the Effects of Various Practice Conditions in Motor Learning,” *Journal of Motor Behavior*, vol. 36, no. 2, pp. 212–224, 2004, doi: [10.3200/jmbr.36.2.212-224](https://doi.org/10.3200/jmbr.36.2.212-224).
- [6] J. R. Anderson, “Acquisition of cognitive skill,” *Psychological Review*, vol. 89, no. 4, pp. 369–406, 1982, doi: [10.1037/0033-295X.89.4.369](https://doi.org/10.1037/0033-295X.89.4.369).
- [7] K. M. Newell, “Motor Skill Acquisition,” *Annual Review of Psychology*, vol. 42, no. 1, pp. 213–237, 1991, doi: [10.1146/annurev.ps.42.020191.001241](https://doi.org/10.1146/annurev.ps.42.020191.001241).
- [8] K. A. Ericsson, R. T. Krampe, and C. Tesch-Römer, “The role of deliberate practice in the acquisition of expert performance,” *Psychological Review*, vol. 100, no. 3, pp. 363–406, 1993, doi: [10.1037/0033-295X.100.3.363](https://doi.org/10.1037/0033-295X.100.3.363).
- [9] G. Wulf and C. H. Shea, “Principles derived from the study of simple skills do not generalize to complex skill learning,” *Psychonomic Bulletin & Review*, vol. 9, no. 2, pp. 185–211, 2002, doi: [10.3758/BF03196276](https://doi.org/10.3758/BF03196276).
- [10] R. Sigrist, G. Rauter, R. Riener, and P. Wolf, “Augmented visual, auditory, haptic, and multimodal feedback in motor learning: A review,” *Psychonomic Bulletin & Review*, vol. 20, no. 1, pp. 21–53, 2013, doi: [10.3758/s13423-012-0333-8](https://doi.org/10.3758/s13423-012-0333-8).
- [11] D. Wood, J. S. Bruner, and G. Ross, “The role of tutoring in problem solving,” *Journal of Child Psychology and Psychiatry*, vol. 17, no. 2, pp. 89–100, 1976, doi: [10.1111/j.1469-7610.1976.tb00381.x](https://doi.org/10.1111/j.1469-7610.1976.tb00381.x).
- [12] J. M. Carroll and C. Carrithers, “Training wheels in a user interface,” *Communications of the ACM*, vol. 27, no. 8, pp. 800–806, 1984, doi: [10.1145/358198.358218](https://doi.org/10.1145/358198.358218).
- [13] L. Bainbridge, “Ironies of automation,” *Automatica*, vol. 19, no. 6, pp. 775–779, 1983, doi: [10.1016/0005-1098\(83\)90046-8](https://doi.org/10.1016/0005-1098(83)90046-8).
- [14] H. Pirsiavash, C. Vondrick, and A. Torralba, “Assessing the Quality of Actions,” in *Computer Vision – ECCV 2014*, 2014, pp. 556–571, doi: [10.1007/978-3-319-10599-4_36](https://doi.org/10.1007/978-3-319-10599-4_36).
- [15] A. Zia and I. Essa, “Automated surgical skill assessment in RMIS training,” *International Journal of Computer Assisted Radiology and Surgery*, vol. 13, no. 5, pp. 731–739, 2018, doi: [10.1007/s11548-018-1735-5](https://doi.org/10.1007/s11548-018-1735-5).
- [16] I. Funke, S. T. Mees, J. Weitz, and S. Speidel, “Video-based surgical skill assessment using 3D convolutional neural networks,” *International Journal of Computer Assisted Radiology and Surgery*, vol. 14, no. 7, pp. 1217–1225, 2019, doi: [10.1007/s11548-019-01995-1](https://doi.org/10.1007/s11548-019-01995-1).
- [17] I. Chikere, R. Oyelaran, and C. Beng, “Ready by the Light, Not by the Hand: What Bakery Trainers Say an Apprentice’s Dough Sense Loses Once a Proofing Cabinet Starts Calling It,” *Slop University technical report*, 2026, doi: [10.5555/slop.86dvpi](https://doi.org/10.5555/slop.86dvpi).
- [18] J. R. Landis and G. G. Koch, “The Measurement of Observer Agreement for Categorical Data,” *Biometrics*, vol. 33, no. 1, pp. 159–174, 1977, doi: [10.2307/2529310](https://doi.org/10.2307/2529310).
- [19] D. Okafor and T. Solberg, “Same Flip, Six Readings: What Actually Trips the Library Silent Floor’s Noise Light Red,” *Slop University technical report*, 2026, doi: [10.5555/slop.bmm3u3](https://doi.org/10.5555/slop.bmm3u3).